Intersatellite LED-Based Optical Wireless Communications for CubeSats

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This study presents a detailed performance analysis of light-emitting diode (LED)-based optical wireless communication (OWC) systems for intersatellite communication (ISC) in CubeSat constellations. Motivated by the stringent size, mass, and power constraints of small satellites, we investigate realistic system limitations, including LED modulation bandwidth, optical beam divergence, and environmental background noise from solar, lunar, and terrestrial sources. Analytical and simulation-based bit error rate models are developed for multiple modulation schemes, namely ON–OFF keying (OOK), M-ary pulse position modulation (M-PPM), multiband carrierless amplitude and phase (m-CAP) modulation, and direct current-biased optical orthogonal frequency-division multiplexing (DCO-OFDM). The proposed models consistently embed LED bandwidth limitations, intersymbol interference, and realistic background-induced shot noise into all four modulation formats, and are validated against independent waveform-level Monte Carlo simulations. The results identify optimal operating regimes for each scheme based on data rate and power constraints, highlighting m-CAP as a strong candidate for high-throughput links and M-PPM for ultralow-power telemetry. In particular, we quantify nontrivial design boundaries in terms of required E_b/N_0 and transmit power versus data rate, range, and background conditions, and show that, even under an intentionally optimistic avalanche photodiode model, PIN receivers remain more power-efficient for the considered CubeSat LED links. Additionally, we quantify the influence of beamwidth and receiver field-of-view on alignment sensitivity and noise resilience. Our findings demonstrate that LED-based OWC can provide scalable, energy-efficient and spectrally agile ISC links, enabling robust communication within next-generation CubeSat missions.

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I. INTRODUCTION

Intersatellite communications (ISC) are essential for the effective operation of low Earth orbit (LEO) satellite systems, which support a wide range of critical applications including global communications, navigation, Earth observation, rendezvous and docking operations, and coordination between LEO and geostationary Earth orbit platforms [1], [2]. The growing interest in distributed space systems and the miniaturization of satellites have driven the deployment of large CubeSat constellations, underscoring the need for robust and power-efficient ISC technologies [2].

CubeSats, characterized by stringent size, mass, and power (SMaP) constraints, present unique design challenges for intersatellite links [3]. Conventional ISC technologies—radio frequency (RF) and laser-based systems—often struggle to meet these constraints due to high power demands, bulky hardware, and complex pointing requirements [4], [5]. In contrast, light-emitting diode (LED)-based optical wireless communication (OWC) systems offer low power consumption, compact integration, and resilience to electromagnetic interference, making them particularly attractive for CubeSat-scale deployments [5], [6], [7].

Since 2010, several studies listed and compared in Table I have explored the feasibility of LED-based OWC for CubeSat ISC, advancing both theoretical modeling and practical validation [8], [9], [10], [11], [12], [13], [14], [15], [16]. Wood et al. [8] proposed the use of LED for ISC for the first time and was analytically evaluated by a simple link budget calculation, in the case of 1 km link at 1.5 Mbps with return-to-zero ON–OFF keying (RZ-OOK) and an avalanche photodiode (APD) as an optical receiver.

Amanor et al. [9], [10], [11], [12], [13] then explored this idea in more depth through several publications. They first confirmed the feasibility of LED-based ISC links by simulation using OOK and multilevel modulation under variable solar background conditions [9], [10], [11]. Specifically, Amanor et al.[9] demonstrated a 0.5 Mbps link over 0.5 km at a bit error rate (BER) of 10^{-6} using nonreturn to zero (NRZ) OOK and a PIN photodiode (PD), while more advanced schemes like M-ary pulse position modulation (M-PPM) and direct current-biased optical orthogonal frequency-division multiplexing (DCO-OFDM) achieved BERs below 10^{-17} in controlled settings. Then, they applied multiobjective optimization to balance LED power, photodetector area, bandwidth, and range—yielding over 3 dB improvement in signal-to-noise ratio (SNR) [11]. Amanor et al. [12] added digital pulse interval modulation (DPIM) to the list of modulations studied, with which they simulated a system achieving 2 Mbps over 0.5 km using a 4 W LED in the C Fraunhofer line (656.28 nm) to mitigate the effect of solar radiations. However, all these works did

TABLE I
Comparison Between the Existing Literature on LED-Based ISC and Our Work (ND: Not Defined)

Reference	Year	Transmitter / Receiver	Modulation	Optical power	Distance	Data rate	BER	Type
[8]	2010	465 nm LED / APD	RZ-OOK	0.117 W	1 km	1.5 Mbps	ND	Simulation
[9]–[12]	2016-2018	656 nm LED / PIN PD with concentrator	NRZ-OOK 8-PPM DPIM	2 W 2 W 4 W	1 km 1 km 0.5 km	0.5 Mbps 0.19 Mbps 2 Mbps	10^{-6} 10^{-9} 10^{-6}	Simulation
[13]	2021	656 nm LED / PIN PD with concentrator	NRZ-OOK with NOMA	10 W	1 km	2 Mbps	10^{-6}	Simulation
[14]	2021	450 nm μLED with focusing lens / SPAD	RZ-OOK	1 mW 463 μW	5.74 km 750 m	100 Mbps 20 Mbps	2×10^{-3} 10^{-5}	Simulation Experiment
[16]	2024	850 nm VCSEL / PIN PD	OOK (MIL-STD-1553)	1.5 mW	1.5 m	1 Mbps	10^{-6}	Experiment
This work	2026	950 nm LED / PIN PD	OOK 4-PPM m-CAP	0.232 W 0.295 W 0.899 W	1 km	1 Mbps 20 Mbps 70 Mbps	10^{-3}	Simulation

not account for LED bandwidth limitations or nonlinearities, and their simplified noise model limited generalization.

At the same time, they evaluated LED-based ISC under solar background noise and line-of-sight (LOS) constraints, proposing nonorthogonal multiple access (NOMA)-based access for cluster-based small satellite networks [13]. They achieved a 2 Mbps aggregate throughput over 1 km, extending results from [17]. More recently, a cooperative NOMA strategy was introduced in [18], improving throughput and energy efficiency under similar constraints. In parallel, improved asymmetrically clipped optical OFDM (ACO-OFDM) and DCO-OFDM were applied in a heterogeneous NanoSat network with optical power control, yielding a spectral efficiency of 3.92 bps/Hz and a BER gain of 3.7 dB [15].

Experimental efforts have also been made, especially, where Griffiths et al. [14] employed μLED transmitters and single-photon avalanche diode (SPAD) receivers to demonstrate OOK-based links up to 100 Mbps, achieving a sensitivity of −55.2 dBm. Experimental tests validated a 750 m link at 20 Mbps, and simulations indicated ranges exceeding 1 km are feasible with 52 dBi directional gain. At the same time, Cossu et al. [16] demonstrated an LED-based system implementing the MIL-STD-1553 protocol to enable point-to-point communication between two nodes on the same satellite at 1 Mbps.

Despite these advances, critical gaps remain. Existing models often neglect key LED-specific constraints such as modulation bandwidth, nonlinearity, and thermal limitations. Moreover, accurate characterization of environmental noise—including solar, lunar, and Earth reflections—and its interplay with hardware limitations is still underexplored. The dynamic nature of orbital configurations also complicates pointing and link availability, particularly in large constellations with diverse orbital planes [5], [6].

This article aims to address these gaps by presenting a comprehensive performance analysis of LED-based ISC links tailored to CubeSat missions. The proposed model incorporates LED angular profiles, bandwidth constraints, and detailed environmental noise sources, including solar, lunar, and terrestrial background light. We evaluate link performance across a range of intersatellite distances, and LED semiangles, offering insight into practical tradeoffs under SMaP constraints.

The core contributions of this article include: 1) the development of a realistic system-level model for LED-based optical ISC links, incorporating practical transmitter and receiver characteristics as well as orbital channel constraints; 2) the formulation of a detailed analytical noise model that integrates solar irradiance, lunar reflection, and Earth albedo under varying LEO conditions; and 3) a comparative performance analysis of multiple modulation schemes—including OOK, PPM, multiband carrierless amplitude and phase (m–CAP) modulation, and DCO–OFDM—adapted to the spectral and nonlinear limitations of LED-based CubeSat platforms. Unlike prior CubeSat LED studies, the analysis explicitly couples LED bandwidth and intersymbol interference (ISI), realistic background-induced shot noise, pointing and channel state information (CSI) efficiencies with new closed-form BER expressions for all four schemes, including new SNR mappings for DCO–OFDM and m–CAP that are consistent with the underlying intensity modulation and direct detection (IM/DD) waveform model. This study thus introduces new BER expressions for these schemes under realistic optical link conditions, addressing key limitations in prior models. From these models, we derive quantitative operating regions—linking data rate, E_b/N_0, and transmit power to range and background conditions—that yield concrete design rules (e.g., when OOK, M-PPM, or m–CAP is actually viable under a 20 MHz LED), and we obtain a nonintuitive detector conclusion showing that PIN receivers remain preferable to APDs even under a deliberately optimistic APD-area assumption. Overall, the proposed framework supports the design of robust, power-efficient communication architectures suitable for next-generation CubeSat constellations operating in diverse and noisy space environments.

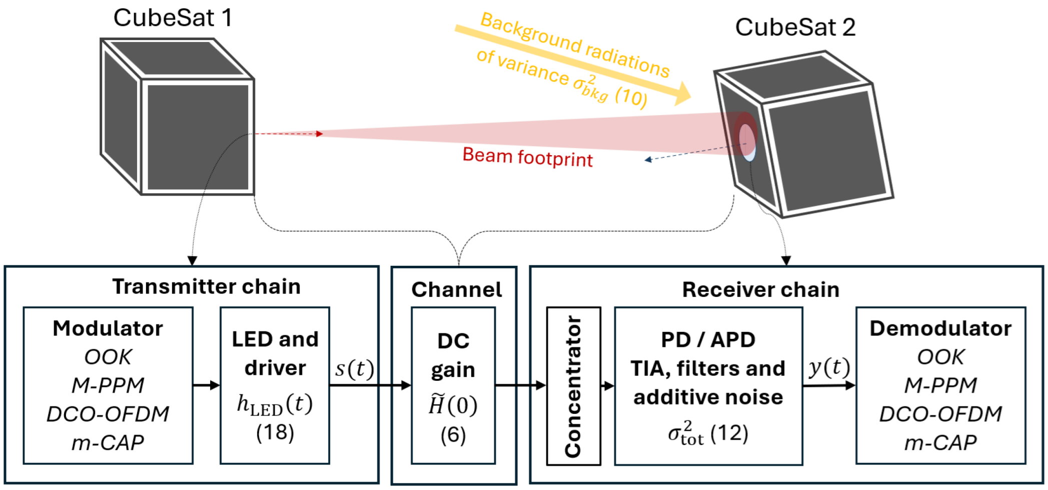


Fig. 1. Illustration of the ISC scenario studied, with possible pointing errors and shifts between CubeSats (illustrated by the partial coverage of the receiver aperture of CubeSat 2, by the beam footprint). At the bottom, the transmitter and receiver chains are detailed, along with the signal flow, with reference to the relevant equations when applicable.

The rest of this article is organized as follows. Section II presents the system and noise models. Section III discusses the modulation schemes and their LED-specific adaptations. Section IV analyzes performance via numerical simulations. Finally, Section V concludes this article and outlines future directions.

II. System Model

This section presents a comprehensive model for LED-based ISC links using LED transmitters and photodetector-based receivers, as illustrated in Fig. 1. The model accounts for the physical structure of the optical transceivers, e.g., the impulse response of the LED h_{LED} which will be further described in Section III-A, the free-space channel between satellites, and the major noise sources in the orbital environment gathered in the total noise variance σ^2_{tot} detailed in Section-II-C. The communication scheme employs IM/DD, supporting modulations such as OOK, PPM, and multicarrier formats like DCO-OFDM or m–CAP modulation [19].

A. Physical Setup and Channel Geometry

The system considered in this work consists of an LED-based transmitter and a photodetector equipped with an optical concentrator and filter. Communication takes place over a line-of-sight (LOS) path between two CubeSats separated by a distance d [20]. The link operates under IM/DD, with the received optical power determined by geometric spreading and the optical front-end characteristics. In line with the existing literature on LED-based ISC, we assume that the LED transmitter has a Lambertian emission pattern, so that its radiant intensity is expressed as [19]

$$I(\theta) = I_0 \cos^m(\theta) \tag{1}$$

where θ denotes the irradiance angle at the transmitter, measured with respect to the LED optical axis, I_0 is the on-axis radiant intensity, and the Lambertian order m is given by

$$m = -\frac{\ln 2}{\ln\cos(\theta_{1/2})} \tag{2}$$

with $\theta_{1/2}$ the LED semiangle at half-power.

The received optical power P_r is related to the transmitted power P_t through

$$P_r = H(0)\,P_t \tag{3}$$

where the LOS channel DC gain $H(0)$ is

$$H(0) = \begin{cases} \frac{(m+1)A_r \cos^m(\theta)\,T_{\mathrm{opt}}}{g}(\psi)\,\cos(\psi)2\pi d^2, & 0 \le \psi \le \psi_c \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Here, A_r is the detector area, T_{opt} the optical front-end transmission coefficient, $g(\psi)$ the concentrator gain, and ψ denotes the incidence angle at the receiver, measured with respect to the photodetector normal, while ψ_c is the receiver field-of-view (FOV) limit. For a hemispherical concentrator with refractive index n_c

$$g(\psi) = \begin{cases} \frac{n_c^2}{\sin^2(\psi_c)}, & 0 \le \psi \le \psi_c \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

On a CubeSat platform, the attitude determination and control system (ADCS) maintains the LOS geometry close to boresight, while residual jitter and slow orbital drift introduce small pointing offsets, as illustrated in Fig. 1. These variations are slow compared to the duration of a data frame, so the geometry can be treated as quasi-static for analytical purposes.

To represent the average loss due to residual pointing offsets without introducing additional random variables in the derivations, we use a deterministic pointing-efficiency factor $\eta_p \in (0, 1]$ that scales the nominal channel gain as $\eta_p H(0)$. Typical values in the range 0.8–1 are consistent with the pointing accuracy of current CubeSat ADCS units and reported performance in LED-based space-link experiments.

Similarly, the accuracy of pilot-based channel estimation is summarized by a CSI efficiency $\eta_{\mathrm{CSI}} \in (0, 1]$, which captures both estimation noise and mild outdatedness between pilot and payload symbols. These two scalar efficiencies are used throughout the analysis to account, in a compact way, for pointing and estimation imperfections on top of the deterministic channel model by scaling the DC gain $H(0)$ into the effective LOS gain $\tilde{H}(0)$

$$\tilde{H}(0) = \sqrt{\eta_{\mathrm{CSI}}}\eta_p H(0). \tag{6}$$

The Doppler shift associated with orbital motion is negligible in the considered IM/DD system [8], [9], [10], [11], [12]. For relative velocities v near 7.5 km/s, the fractional wavelength change is $v/c \approx 2.5 \times 10^{-5}$, corresponding to a shift of about 0.024 at 950 nm, which is much smaller than typical optical filter bandwidths. Doppler effects are therefore ignored in the sequel.

B. Signal Representation

Under IM/DD operation, the transmitted information is conveyed through the instantaneous optical intensity. At the receiver, the output $y(t)$ of the photodetector is modeled as [19]

$$y(t) = R\tilde{H}(0)x(t) + n(t) \tag{7}$$

where $x(t) \geq 0$ denotes the transmitted optical power, $\tilde{H}(0)$ is the effective LOS channel gain including pointing efficiency, R is the detector responsivity, and $n(t)$ accounts for thermal, background-induced, and signal-dependent shot noise [19], [21]. The quantity $y(t)$ corresponds to the photocurrent at the receiver front-end.

Channel estimation is performed once per frame using embedded pilot symbols. Since the LOS gain evolves slowly over the frame duration, the resulting estimation quality can be summarized through a CSI efficiency factor $\eta_{\mathrm{CSI}} \in (0, 1]$. Combined with the pointing efficiency η_p, the postdetection SNR that enters all BER expressions is written as

$$\mathrm{SNR}_{\text{p-CSI}} = \eta_p^2\, \eta_{\mathrm{CSI}}\, \mathrm{SNR} \tag{8}$$

where SNR denotes the SNR under perfect alignment and perfect CSI. This representation preserves analytical clarity while enabling a direct mapping between practical spacecraft performance and the predicted BER.

The photodetector significantly affects both responsivity and noise characteristics. CubeSat receivers typically employ either PIN PDs or APDs. PIN PDs offer moderate responsivity ($R_{\mathrm{PIN}} = 0.4-0.6$ A/W) with minimal excess noise, making them suitable for low-power architectures. APDs provide internal gain ($G \approx 10-100$), modeled as $R_{\mathrm{APD}} = G\,R_{\mathrm{PIN}}$, but introduce additional noise captured by an excess noise factor $F > 1$ [22], [23]. The impact of these architectures on system sensitivity is examined in Section IV.

C. Noise and Background Illumination Modeling

In LED-based ISC links, the received signal is affected by multiple noise sources that impact system sensitivity and reliability. These sources include thermal noise from receiver electronics, background-induced shot noise resulting from ambient illumination, and signal-dependent shot noise associated with the received optical power. Together, these components determine the total noise variance observed at the receiver [19].

Thermal noise originates from the random motion of charge carriers within the receiver's front-end components. For a system with load resistance R_L, electrical bandwidth B, and equivalent noise temperature T_e, the thermal noise variance is expressed as [24]

$$\sigma_{\mathrm{th}}^2 = \frac{4k_B T_e B}{R_L} \tag{9}$$

where k_B is Boltzmann's constant.

Shot noise arises due to the discrete nature of photon detection and consists of two contributions. The first corresponds to ambient illumination—primarily sunlight and secondary reflections from celestial bodies—and is captured by the background-induced shot noise variance [19], [20]

$$\sigma_{\mathrm{bkg}}^2 = 2qRFBP_{\mathrm{bkg}} \tag{10}$$

where q is the elementary charge, R is the responsivity of the photodetector, F is the APD excess-noise factor (equal to 1 for PIN PDs), and P_{bkg} is the incident background optical power.

The second contribution corresponds to the signal-dependent shot noise, which depends on the received optical power P_r from the transmitting satellite. This is given by [24]

$$\sigma_{\mathrm{shot}}^2 = 2qRFBP_r. \tag{11}$$

The total input-referred noise variance at the receiver is the sum of all three components

$$\sigma_{\mathrm{tot}}^2 = \sigma_{\mathrm{th}}^2 + \sigma_{\mathrm{bkg}}^2 + \sigma_{\mathrm{shot}}^2. \tag{12}$$

In this work, the same physical noise model is used as in standard IM/DD free space optical communication receivers, but all parameters are evaluated for the intersatellite environment. In particular, the background power P_{bkg} is computed from extraterrestrial irradiance and orbital geometry rather than from terrestrial ambient light.

To accurately evaluate P_{bkg}, we consider direct solar irradiance and reflected contributions from the Moon and Earth. The Sun is modeled as a blackbody radiator at temperature T, and its spectral radiance $W(\lambda, T)$ at wavelength λ is described by Planck's law [10], [18]

$$W(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1} \tag{13}$$

where h is Planck's constant and c is the speed of light. We validate our analytical blackbody model against the ASTM G173–03 empirical dataset, supplemented by the high-resolution reference spectra from Gueymard et al. [25].

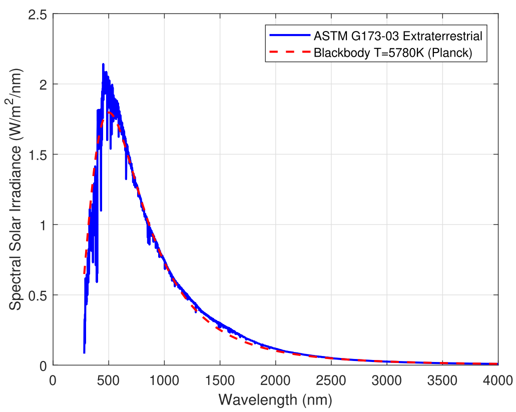


Fig. 2. Comparison between the empirical ASTM G173–03 extraterrestrial solar irradiance and the analytical blackbody model at $T = 5780\,\mathrm{K}$.

As shown in Fig. 2, the analytical Planck curve aligns closely with these empirical standards, confirming the reliability of the radiative model.

The effective solar optical power incident on the detector through an optical filter is computed by integrating this spectral radiance over the receiver's operational bandwidth

$$P_{\text{sun}} = A_r T_{\text{opt}} \Omega_{\text{sun}} \int_{\lambda_a}^{\lambda_b} W(\lambda, T) \cdot \tau_{\text{filter}}(\lambda)\, d\lambda \tag{14}$$

where A_r is the detector area, T_{opt} the optical system transmission coefficient, $\tau_{\text{filter}}(\lambda)$ the spectral transmission profile of the filter, and Ω_{sun} the solid angle subtended by the Sun.

Reflected solar illumination from Earth and the Moon is incorporated using a simplified Lambertian albedo model

$$P_{\text{ref}} = A_r T_{\text{opt}} \left(\frac{\rho_{\text{moon}} \Omega_{\text{moon}} + \rho_{\text{earth}} \Omega_{\text{earth}}}{\pi} \right) \times \int_{\lambda_a}^{\lambda_b} W(\lambda, T) \cdot \tau_{\text{filter}}(\lambda)\, d\lambda \tag{15}$$

where $\rho_{\text{moon}}, \rho_{\text{earth}}$ are their respective average albedos, and $\Omega_{\text{moon}}, \Omega_{\text{earth}}$ are their solid angles at the satellite [26], [27].

Daytime conditions include both direct and reflected illumination, while nighttime includes only reflected contributions

$$P_{\text{bkg}} = \begin{cases} P_{\text{sun}} + P_{\text{ref}}, & \text{daytime} \\ P_{\text{ref}}, & \text{nighttime.} \end{cases} \tag{16}$$

This modeling framework ensures that the background-induced shot noise reflects realistic orbital illumination conditions. Under typical LEO daytime conditions, background-induced shot noise dominates the receiver noise budget, while thermal noise remains comparatively small and is included for completeness.

III. MODULATIONS WITH LED BANDWIDTH LIMITATIONS

In this work, we consider four modulation schemes for the LED-based ISC links: OOK, PPM, DCO–OFDM, and m–CAP. In the following, we describe the signal models, incorporate the effects of the LED's finite bandwidth, and present the theoretical BER expressions used in the analysis. Unless otherwise stated, all SNR quantities are understood to include the pointing and CSI efficiencies introduced in Section-II-B.

A. OOK

OOK is the simplest form of intensity modulation for optical communication, where binary data is conveyed by the presence (logic "1") or absence (logic "0") of a pulse [28], [29]. A logic "1" is transmitted as a rectangular pulse with peak amplitude $i_{\text{peak,OOK}}$ (proportional to the average optical power P_{avg}), while a logic "0" corresponds to no pulse. The transmitted signal can thus be modeled as a train of rectangular pulses, each of duration equal to the bit period T_b. Mathematically, the signal is given by [30]

$$s_{\text{OOK}}(t) = \sum_{n=0}^{N-1} b_n\, i_{\text{peak,OOK}}\, \text{rect}\left(\frac{t - nT_b}{T_b} \right) \tag{17}$$

where $b_n \in \{0, 1\}$ is the nth bit, $\text{rect}(t/T_b)$ equals 1 for $0 \le t < T_b$ and 0 otherwise, and N is the total number of transmitted bits.

This electrical driving current is then converted into optical power through the LED electro–optical efficiency κ [W/A]. However, due to the bandwidth limitation of the LED, $s_{\text{OOK}}(t)$ is also distorted and takes the form $s_{\text{out}}(t)$, as further detailed below. The transmitted optical power is thus $P_t(t) = \kappa\, s_{\text{out}}(t)$. After free-space propagation with DC gain $H(0)$, the received optical power associated with a "1" symbol is $P_r(t) = \kappa\, H(0)\, s_{\text{out}}(t)$, and the corresponding photocurrent at the detector is $i_r(t) = R\, P_r(t)$. This makes the dependence of the received signal on the wireless channel coefficient $H(0)$ explicit in the subsequent SNR and BER expressions.

To explicitly include the effects of geometric loss, pointing inaccuracy, and imperfect CSI in a single term, we use the effective DC channel gain $\tilde{H}(0)$, as defined in (6), which replaces the nominal LOS gain in the link budget and captures all deterministic channel impairments in a unified form.

The LED is modeled as a first-order low-pass filter with impulse response [28]

$$h_{\text{LED}}(t) = \frac{1}{\tau} e^{-t/\tau}\, u(t) \tag{18}$$

where τ is the LED time constant and $u(t)$ is the unit step function. The corresponding frequency response is given by

$$H_{\text{LED}}(f) = \frac{1}{1 + j2\pi f \tau} \tag{19}$$

indicating a single-pole roll-off that limits the LED's modulation bandwidth. Due to this bandwidth limitation, the transmitted pulse is distorted; the output of the LED filter is given by the convolution

$$s_{\text{out}}(t) = \int_0^t s_{\text{OOK}}(\xi)\, h_{\text{LED}}(t - \xi)\, d\xi. \tag{20}$$

To quantify the impact of the LED bandwidth, we define the effective current over one bit period as

$$i_{\text{peak,eff}} = \frac{1}{T_b}\int_0^{T_b} s_{\text{out}}(t)\,dt. \tag{21}$$

By substituting (20) and noting that for $0 \le \tau < T_b$, the transmitted pulse is

$$s_{\text{OOK}}(\tau) = i_{\text{peak,OOK}} \cdot \text{rect}\left(\frac{\tau}{T_b}\right)$$

one can show that

$$i_{\text{peak,eff}} = i_{\text{peak,OOK}}\left[1 - \frac{\tau}{T_b}\left(1 - e^{-T_b/\tau}\right)\right]. \tag{22}$$

Thus, for each "1" symbol the effective current is reduced from the nominal $i_{\text{peak,OOK}}$ by an amount that depends on the ratio τ/T_b; a larger time constant (or a shorter bit duration) results in greater pulse attenuation.

At the receiver, the corresponding average effective photocurrent for a "1" symbol is

$$i_{r,\text{eff}} = R\,\kappa\,\tilde{H}(0)\,i_{\text{peak,eff}}. \tag{23}$$

In addition to this deterministic reduction of the "1" level, the finite LED response produces a residual tail that extends into subsequent bit intervals. For random bit sequences, this tail behaves as an additional fluctuating current superimposed on the decision statistic. Its contribution can be approximated, in terms of shot-noise variance, by

$$\sigma_{\text{ISI}}^2 \approx q\,I_{\text{ISI}}\,R_b. \tag{24}$$

For OOK, the mean ISI current as shown in Appendix A is $I_{\text{ISI}} = \frac{\tau}{T_b}\left(\frac{R\,\kappa\,\tilde{H}(0)\,i_{\text{peak,OOK}}}{2}\right)(1 - e^{-T_b/\tau})$, where I_{ISI} denotes the mean residual LED current originating from preceding symbols and $R_b = 1/T_b$ is the bit rate. The total decision noise variance then becomes

$$\sigma_{\text{tot,eff}}^2 = \sigma_{\text{tot}}^2 + \sigma_{\text{ISI}}^2 \tag{25}$$

with σ_{tot}^2 given by the physical noise model in Section II-C. We also introduce the normalized ISI factor $\delta_{\text{ISI}} \triangleq \sigma_{\text{ISI}}^2/\sigma_{\text{tot}}^2$, so that $\sigma_{\text{tot,eff}}^2 = \sigma_{\text{tot}}^2(1 + \delta_{\text{ISI}})$.

We define the nominal SNR (SNR_0), corresponding to the ratio E_b/N_0 of the energy per bit E_b to the noise power spectral density N_0, as

$$\frac{E_b}{N_0} \triangleq \text{SNR}_0 = \frac{\left(R\,\kappa\,\tilde{H}(0)\,i_{\text{peak,OOK}}\right)^2}{4\,\sigma_{\text{tot}}^2} \tag{26}$$

which corresponds to the ideal OOK case without LED distortion and in the absence of ISI (rectangular pulses, memoryless LED). The associated ISI-aware nominal SNR is then

$$\text{SNR}_{\text{eff,OOK}} = \frac{\left(R\,\kappa\,\tilde{H}(0)\,i_{\text{peak,OOK}}\right)^2}{4\,\sigma_{\text{tot,eff}}^2} = \frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}. \tag{27}$$

In practice, the LED's bandwidth reduces the effective signal current. To account for partial pulse settling in the "1" case, and as further detailed in Appendix A, the approximate BER under standard threshold detection is expressed as

$$P_{be,\text{OOK}}^{\text{eff}} = \frac{1}{2}Q\left(\sqrt{\frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}}\right) + \frac{1}{2}Q\left(2\sqrt{\frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}}\left[\frac{1}{2} - \frac{\tau}{T_b}\left(1 - e^{-T_b/\tau}\right)\right]\right) \tag{28}$$

where $Q(x) = \frac{1}{2}\,\text{erfc}(x/\sqrt{2})$. The first term represents the error probability when a "0" is transmitted, while the second term reflects the degradation in the "1" symbol due to the incomplete LED response [30], [31].

The closed-form expressions for the effective peak current (22), the ISI-aware SNR (27), and the corresponding BER (28) provide a compact performance model that includes both deterministic LED distortion and the dominant ISI contribution, while explicitly embedding the optical channel coefficient $H(0)$ in the link budget.

B. M-PPM

M-PPM encodes data by placing a single pulse within one of $L_{\text{avg}} = 2^M$ time slots in each symbol period, where M is the number of bits per symbol. The transmitted signal is expressed as [30]

$$s_{\text{PPM}}(t) = \sum_{k=1}^{K} a_k\,p\big(t - kT_s\big) \tag{29}$$

where a_k is equal to the peak current $i_{\text{peak,PPM}}$ in the active slot and zero otherwise, $p(t)$ denotes the pulse shape, T_s is the PPM slot duration, and K is the number of transmitted symbols.

Since only one slot contains a pulse and the others are zero, PPM is energy efficient. However, in LED-based systems, the narrow modulation bandwidth introduces pulse spreading, which distorts the transmitted waveform and reduces the peak amplitude at the detector.

To evaluate this effect, we approximate the shape of the bandwidth-limited PPM pulse using the same expression derived for OOK in the presence of LED filtering. Specifically, we use the effective average current (21) from the OOK derivation, scaled to the PPM slot duration T_s, to compute an equivalent effective SNR. This allows reuse of the analytical BER expressions originally derived for ideal rectangular pulses by adjusting them to account for the reduced pulse amplitude due to LED bandwidth limitations.

Analogously to the OOK case, the bandwidth-limited current over one PPM slot can be written as

$$i_{\text{peak,eff,PPM}} = i_{\text{peak,PPM}}\left[1 - \frac{\tau}{T_s}\left(1 - e^{-T_s/\tau}\right)\right] \tag{30}$$

and the corresponding effective electrical SNR per active slot, including the physical noise components and the small ISI contribution between adjacent slots, is denoted by $\text{SNR}_{\text{eff,PPM}}$. In the normalized analysis that follows, $\text{SNR}_{\text{eff,PPM}}$ is obtained from the same link budget as OOK, i.e., it embeds the factor $R\kappa H(0)$ and the pointing/CSI efficiencies introduced in Section II-B.

Accordingly, the effective hard-decision symbol error probability is approximated by

$$P_{se,\mathrm{PPM}}^{\mathrm{hard,eff}} = \frac{L_{\mathrm{avg}}-1}{L_{\mathrm{avg}}} Q\left(\sqrt{ML_{\mathrm{avg}}\frac{\mathrm{SNR}_{\mathrm{eff,PPM}}}{2}}\right) + \frac{1}{L_{\mathrm{avg}}} Q\left(2\sqrt{ML_{\mathrm{avg}}\frac{\mathrm{SNR}_{\mathrm{eff,PPM}}}{2}} \times \left[\frac{1}{2} - \frac{\tau}{T_s}\left(1-e^{-T_s/\tau}\right)\right]\right) \quad (31)$$

where $\mathrm{SNR}_{\mathrm{eff,PPM}}$ is evaluated using $i_{\mathrm{peak,eff,PPM}}$ and the total noise variance in the same way as for OOK.

The overall symbol error probability is given by [28]

$$P_{sy,\mathrm{PPM}}^{\mathrm{hard,eff}} = 1 - \left[1 - P_{se,\mathrm{PPM}}^{\mathrm{hard,eff}}\right]^{L_{\mathrm{avg}}} \quad (32)$$

and the corresponding bit error rate is

$$P_{be,\mathrm{PPM}}^{\mathrm{hard,eff}} = \frac{L_{\mathrm{avg}}/2}{L_{\mathrm{avg}}-1} P_{sy,\mathrm{PPM}}^{\mathrm{hard,eff}}. \quad (33)$$

C. DCO–OFDM

DCO–OFDM is a widely adopted multicarrier modulation technique for IM/DD optical systems. It modulates digital data onto multiple orthogonal subcarriers using an inverse fast Fourier transform (IFFT) applied to a Hermitian-symmetric quadrature amplitude modulation (QAM) symbol vector, ensuring a real-valued time-domain signal [32]. To meet the nonnegativity constraint of IM/DD systems, a DC bias is added, and any residual negative values are clipped.

In this work, the IFFT output is explicitly modeled as a Gaussian-distributed sequence $x[n]$ (Appendix B), enabling a consistent treatment of DC bias and clipping through the Bussgang decomposition. The discrete-time transmitted signal is then given by

$$s_{\mathrm{DCO}}[n] = \max\{x[n] + B_{\mathrm{DC}},\ 0\} \quad (34)$$

with

$$x[n] = \frac{1}{N_{\mathrm{FFT}}} \sum_{k=0}^{N_{\mathrm{FFT}}-1} X_k\, e^{j2\pi kn/N_{\mathrm{FFT}}} \quad (35)$$

where X_k denotes the Hermitian-symmetric QAM symbols and N_{FFT} the IFFT size. For the large number of subcarriers used here, $x[n]$ closely follows a zero-mean Gaussian distribution, allowing the clipping model to be expressed using the parameters P_{sig} and P_{clip} derived in Appendix B.

The addition of the DC bias and subsequent clipping introduce nonlinear distortion, while further performance degradation arises from cyclic prefix (CP) overhead and the exclusion of the DC and Nyquist subcarriers.

Using the Bussgang decomposition, the total average electrical power at the LED driver is written as $P_{\mathrm{avg}} = P_{\mathrm{sig}} + P_{\mathrm{clip}}$, where P_{sig} is the power of the useful linear component and P_{clip} represents uncorrelated distortion from clipping [33]. This decomposition enables a consistent link between the nominal OOK-based SNR and the DCO–OFDM effective SNR.

To quantify these effects, the effective SNR is derived from the nominal OOK-based SNR as

$$\mathrm{SNR}_{\mathrm{eff,DCO}} = \mathrm{SNR}_0 \cdot \frac{P_{\mathrm{sig}}}{P_{\mathrm{avg}}}\left[1 + \frac{P_{\mathrm{clip}}}{P_{\mathrm{avg}}}\mathrm{SNR}_0\right]^{-1} \cdot \frac{N_{\mathrm{FFT}}-2}{N_{\mathrm{FFT}}+N_{\mathrm{cp}}} \cdot \frac{1}{\log_2 M_{\mathrm{QAM}}} \quad (36)$$

where $P_{\mathrm{sig}}/P_{\mathrm{avg}}$ and $P_{\mathrm{clip}}/P_{\mathrm{avg}}$ are computed from the Gaussian clipping model (Appendix B). The terms in (36), respectively, capture useful signal power, clipping-induced distortion, CP and inactive-subcarrier overhead, and the conversion from symbol-level to bit-level SNR.

For the operating regime considered in this study (moderate DC bias and negligible upper clipping), the Gaussian clipping model simplifies to $P_{\mathrm{sig}}/P_{\mathrm{avg}} \approx 1/(1+\gamma)$ with $\gamma = B_{\mathrm{DC}}^2/\sigma_x^2$, leading to the compact form

$$\mathrm{SNR}_{\mathrm{eff,DCO}} \approx \frac{\mathrm{SNR}_0}{1+\gamma} \cdot \frac{N_{\mathrm{FFT}}-2}{N_{\mathrm{FFT}}+N_{\mathrm{cp}}} \cdot \frac{1}{\log_2 M_{\mathrm{QAM}}}. \quad (37)$$

Each subcarrier experiences further attenuation due to the LED's low-pass response

$$\mathrm{SNR}_{\mathrm{eff,DCO}}^{(n)} = \mathrm{SNR}_{\mathrm{eff,DCO}} \cdot \left|H_{\mathrm{LED}}\left(f_{\mathrm{DCO}}^{(n)}\right)\right|^2. \quad (38)$$

Here $f_{\mathrm{DCO}}^{(n)} = n\,\Delta f$ is the center frequency of the nth subcarrier, and

$$\Delta f = \frac{f_{\mathrm{samp}}}{N_{\mathrm{FFT}}} \quad (39)$$

is the subcarrier spacing, with f_{samp} being the sampling rate.

Assuming Gray-coded QAM, the BER on the nth subcarrier is approximated as

$$P_{be}^{\mathrm{DCO}}(n) = \frac{4}{\log_2 M_{\mathrm{QAM}}}\left(1 - \frac{1}{\sqrt{M_{\mathrm{QAM}}}}\right) \times Q\left(\sqrt{\frac{3\,\log_2 M_{\mathrm{QAM}}\,\mathrm{SNR}_{\mathrm{eff,DCO}}^{(n)}}{M_{\mathrm{QAM}}-1}}\right). \quad (40)$$

Finally, the number of usable subcarriers is $N_{\mathrm{sub}} = N_{\mathrm{FFT}}/2 - 1$, accounting for the removal of the DC component and the Nyquist frequency to maintain Hermitian symmetry and real-valued output. The overall BER is therefore obtained by averaging over all active subcarriers

$$P_{be}^{\mathrm{DCO}} = \frac{1}{N_{\mathrm{sub}}} \sum_{n=1}^{N_{\mathrm{sub}}} P_{be}^{\mathrm{DCO}}(n). \quad (41)$$

D. m–CAP

m–CAP enhances spectral efficiency by partitioning the available bandwidth into N_{CAP} orthogonal subbands, each modulated independently using QAM [34], [35]. This structure reduces ISI without the need for explicit subcarrier oscillators, making it attractive for bandwidth-limited optical systems. Compared to DCO–OFDM, which employs a Hermitian-symmetric IFFT to generate real-valued multicarrier waveforms, m–CAP uses orthogonal pulse-shaping filters for each subband, with center frequencies chosen to tile the usable LED bandwidth.

Each subband carries a data stream filtered by a square-root raised cosine (SRRC) pulse-shaping filter with roll-off factor β. The center frequency of the nth subband is given by

$$f_{\mathrm{CAP}}^{(n)} = \frac{(2n-1)\,R_b}{\log_2(M_{\mathrm{QAM}})\,N_{\mathrm{CAP}}} \cdot \frac{1+\beta}{2}, \quad n = 1, \ldots, N_{\mathrm{CAP}} \tag{42}$$

where R_b is the overall bit rate, M_{QAM} the constellation size, and N_{CAP} the number of subbands.

After subband pulse shaping and summation, the composite time-domain signal is DC biased to satisfy the IM/DD nonnegativity constraint. As in DCO–OFDM, the m–CAP waveform can be written in the form of (34), with $x[n]$ representing the pulse-shaped multiband signal. However, because pulse shaping produces smoother amplitude statistics than OFDM, m–CAP exhibits a lower peak-to-average power ratio (PAPR) and therefore requires less DC bias than DCO–OFDM to ensure nonnegativity.

To model the bias penalty consistently with Section III-C, we define the normalized bias ratio $\gamma = B_{\mathrm{DC}}^2/\sigma_x^2$, where σ_x^2 is the variance of the composite m–CAP waveform. This captures the reduction in useful electrical power caused by the DC bias. Pulse shaping also introduces an excess-bandwidth factor: larger roll-off β increases bandwidth but reduces the effective energy per bit. Combining both effects, the effective SNR of m–CAP is approximated as

$$\mathrm{SNR}_{\mathrm{eff,CAP}} = \frac{\mathrm{SNR}_0}{1+\gamma} \cdot \frac{\log_2(M_{\mathrm{QAM}})}{1+\beta} \tag{43}$$

where SNR_0 is the nominal OOK-based SNR defined in the system model.

Each subband experiences different attenuation due to the LED's frequency response

$$\mathrm{SNR}_{\mathrm{eff,CAP}}^{(n)} = \mathrm{SNR}_{\mathrm{eff,CAP}} \cdot \left| H_{\mathrm{LED}}(f_{\mathrm{CAP}}^{(n)}) \right|^2. \tag{44}$$

Assuming Gray-coded QAM, the BER on the nth subband is approximated as

$$P_{be}^{\mathrm{CAP}}(n) = \frac{4}{\log_2(M_{\mathrm{QAM}})} \left(1 - \frac{1}{\sqrt{M_{\mathrm{QAM}}}}\right) \times Q\left(\sqrt{\frac{3\,\log_2(M_{\mathrm{QAM}})\,\mathrm{SNR}_{\mathrm{eff,CAP}}^{(n)}}{M_{\mathrm{QAM}}-1}}\right) \tag{45}$$

and the overall BER is

$$P_{be}^{\mathrm{CAP}} = \frac{1}{N_{\mathrm{CAP}}} \sum_{n=1}^{N_{\mathrm{CAP}}} P_{be}^{\mathrm{CAP}}(n). \tag{46}$$

This structure enables m–CAP to offer a flexible tradeoff between bandwidth efficiency, implementation complexity, and required transmit power, making it a strong candidate for LED-based intersatellite links [36].

IV. NUMERICAL RESULTS AND DISCUSSION

A. Simulation Methodology and Parameters

In this section, we present simulation results evaluating the performance of various modulation schemes for LED-based intersatellite OWC between CubeSats. The analysis focuses on key performance metrics such as BER, E_b/N_0, data rates (R_b), required transmit power (P_t), and the impact of background noise under different conditions (daytime and nighttime). All Monte Carlo results are obtained from waveform-level simulations that are independent of the analytical BER formulas. For each modulation scheme, random bits are generated, mapped to symbols, converted into continuous-time waveforms, passed through the LED low-pass model and the physical noise processes of Section II-C, and then demodulated to count bit errors. The Q-function-based BER expressions of Section III are evaluated only outside the Monte Carlo loop, for plotting and comparison. The agreement between curves therefore validates the analytical approximations against an independent physical simulation, rather than reimplementing the same equations in discrete time. Experimental validation with hardware-in-the-loop is left as an important direction for future work.

Unless stated otherwise, the simulation parameters used in our study are summarized in Table II. Note that the pointing and CSI efficiencies are such that $\eta_p = 0.8$ and $\eta_{\mathrm{CSI}} = 0.9$, which are values consistent with the CubeSat jitter and channel-estimation conditions analyzed in another ongoing work, and are thus used here to capture realistic performance degradation in the evaluated link budget.

B. BER Performance of Modulation Schemes at Different Data Rates

Fig. 3 presents the BER performance of four modulation schemes—OOK, M-PPM, DCO-OFDM, and m-CAP—evaluated at data rates of 15 and 60 Mbps. The BER is plotted as a function of E_b/N_0, using both analytical models (solid lines) and Monte Carlo simulations (markers), based on the expressions derived in Section III. Monte Carlo simulations were carried out by transmitting 10^6 randomly generated bits per scenario. The LED's bandwidth constraint was modeled with a first-order low-pass filter of cutoff frequency 20 MHz, and its half-power was set to $\theta_{1/2} = 5^\circ$, whereas the PIN receiver's FOV was set to $\psi_c = 5^\circ$, unless otherwise stated.

First of all, the strong agreement between simulated and analytical results confirms the accuracy of our theoretical derivations. Then, at 15 Mbps [see Fig. 3(a)], M-PPM achieves the best performance, reaching a BER of 10^{-3} at $E_b/N_0 \approx 10$ dB. OOK and m-CAP require slightly higher energy (approximately 12 and 13 dB, respectively), while DCO-OFDM reaches the same BER at about 15 dB. The BER threshold of 10^{-3} is a common target in OWC systems, providing a balance between link quality and error correction overhead [28].

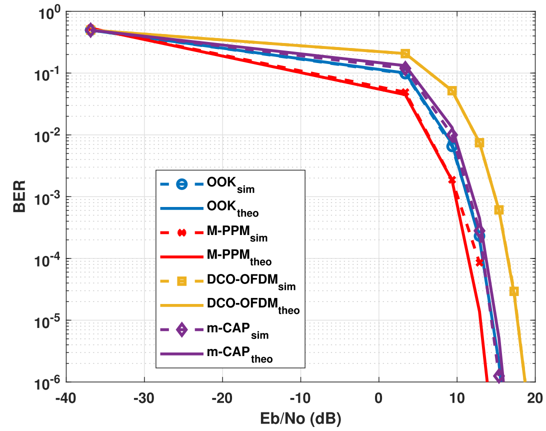
At 60 Mbps [see Fig. 3(b)], M-PPM is no longer viable, with a BER plateauing near 0.5 due to pulse durations

TABLE II
Simulation Parameters

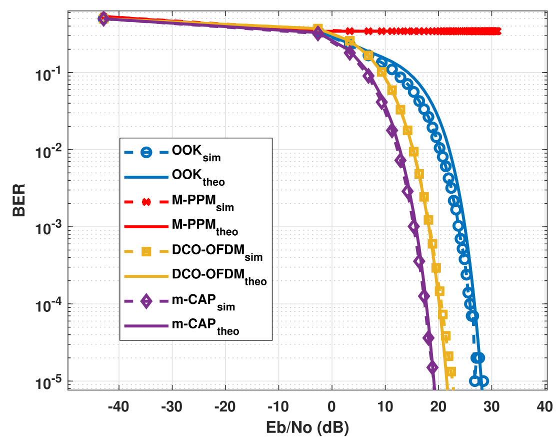
Parameter	Value
System Setup	
Transmit power P_t	0–1 W
Communication distance d	100–2000 m
Data rate R_b	1, 20, 70 Mbps
LED wavelength λ	950 nm
LED semi-angle $\theta_{1/2}$	5°, 10°, 20°
LED bandwidth (3-dB)	20 MHz
B_{dB} (DCO-OFDM, m-CAP)	7 dB
Receiver Configuration	
Receiver FOV ψ_c	5°, 10°, 20°
Photodetector area A_r	$1\,\text{cm}^2$ (upper bound)
Photodetector types	PIN (R_{PIN} = 0.65 A/W) and APD (R_{APD} = 6.5 A/W)
Concentrator refractive index n_c	1.5
Filter bandwidth $\Delta\lambda$	10 nm
Filter transmission τ_{filter}	90%
Optical transmission T_{opt}	90%
Excess noise factor F	1 (PIN), 3 (APD)
Load resistance R_L	1 kΩ
Receiver temperature T_e	300 K
Dark current I_d	Neglected
Environmental Conditions	
Ambient irradiance P_{bkg}	11.8 W/m² (day), 2.0 W/m² (night)
Solar temperature	5780 K
Earth/Moon albedo	0.3 / 0.12
Spectral bounds λ_a–λ_b	945–955 nm
Solid angles $\Omega_{\text{sun}}, \Omega_{\text{moon}}, \Omega_{\text{earth}}$	Computed geometrically
Pointing efficiency η_p	0.8
CSI efficiency η_{CSI}	0.9
Modulation Parameters	
Modulation schemes	OOK, 4-PPM, DCO-OFDM, m-CAP
Target BER	10^{-3}
Monte Carlo size	10^6 bits
DCO-OFDM parameters	$N_{\text{FFT}} = 64$, $N_{\text{CP}} = 4$, QAM-4
m-CAP parameters	$N_{\text{sub}} = 8$, $\beta = 0.1$, QAM-4, SRRC filter

becoming too short to be faithfully reproduced by the LED, leading to severe ISI. The other modulation schemes remain functional: m-CAP offers the best performance, achieving 10^{-3} at approximately 16 dB, followed by DCO-OFDM at 18 dB and OOK at 25 dB. m-CAP's superior performance stems from its subband filtering and pulse shaping, which mitigate bandwidth limitations more effectively than single-carrier schemes, as well as its reduced PAPR compared to DCO-OFDM.

The degradation observed when increasing the data rate is consistent with the analytical rate—BER coupling derived in Section III. For single-carrier schemes (OOK and M-PPM), the effective signal amplitude scales with the LED settling factor $[1 - \tau R_b(1 - e^{-1/(\tau R_b)})]$, while the receiver noise variance grows approximately linearly with the noise bandwidth $B \propto R_b$. Thus, higher R_b simultaneously reduces the useful optical current and increases the total noise, leading to a rightward shift of the BER curves. For multicarrier schemes (DCO-OFDM and m-CAP), larger R_b values push more subcarriers or subbands into the LED's roll-off region, where $|H_{\text{LED}}(f)|^2 < 1$, lowering the effective per-subcarrier SNR. This analytical dependence explains the steeper E_b/N_0 requirements at $R_b = 60$ Mbps compared to 15 Mbps, as seen in Fig. 3.



(a)



(b)

Fig. 3. BER performance of OOK, M-PPM, DCO-OFDM, and m-CAP at (a) 15 Mbps and (b) 60 Mbps. (a) $R_b = 15$ Mbps. (b) $R_b = 60$ Mbps.

Overall, M-PPM is highly energy-efficient at low data rates but severely bandwidth-limited beyond roughly 15 Mbps. OOK remains simple and robust but becomes power-inefficient at higher throughputs. DCO-OFDM and m-CAP exhibit better scalability, with m-CAP providing the most favorable balance between spectral efficiency and BER performance. These behaviors are further quantified in Section IV-C, which analyzes the required E_b/N_0 versus data rate.

C. Required E_b/N_0 Versus Data Rate

To further analyze the influence of data rate R_b on modulation scheme performance, we evaluate the required E_b/N_0 to achieve a target BER of 10^{-3}, as shown in Fig. 4.

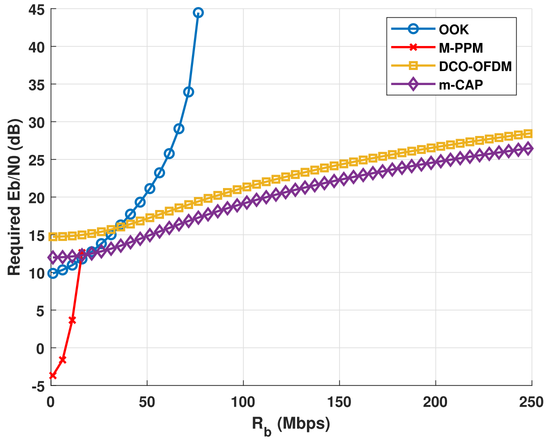


Fig. 4. Required E_b/N_0 versus data rate to achieve a BER of 10^{-3} for OOK, M-PPM, DCO-OFDM, and m-CAP.

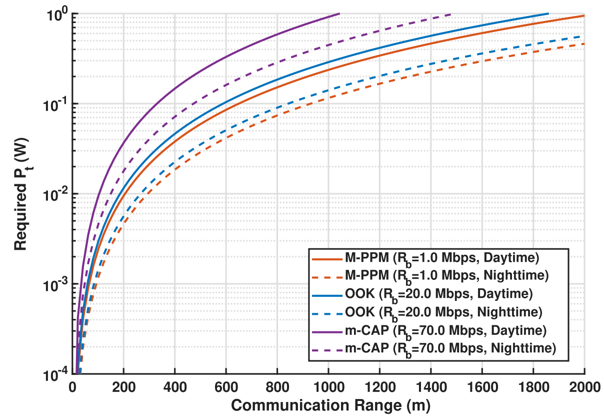


Fig. 5. Required transmit power P_t versus communication range under daytime and nighttime background noise, for M-PPM (1 Mbps), OOK (20 Mbps), and m-CAP (70 Mbps).

The results are derived from analytical models for OOK, M-PPM, DCO-OFDM, and m-CAP, described in Section III.

M-PPM is the most energy-efficient scheme at very low data rates. It achieves the target BER with very low E_b/N_0 values (as low as -3.7 dB at 1 Mbps) and remains favorable up to around 15–16 Mbps. However, due to the LED's limited bandwidth and its inability to transmit increasingly short pulses, the required E_b/N_0 rises sharply beyond this point. M-PPM becomes impractical above 16 Mbps, where ISI significantly degrades performance.

OOK exhibits a smoother increase in required E_b/N_0 with data rate. It is not optimal at low data rates, where M-PPM performs better, but becomes the most efficient option in the intermediate range between 16 and 20 Mbps. Beyond 20 Mbps, however, OOK's energy demands rise more steeply than those of the other schemes. At around 76 Mbps, OOK requires over 44 dB to maintain the target BER, and beyond this point, it becomes unusable under the given system conditions.

m-CAP consistently outperforms DCO-OFDM across the entire range of data rates. Both schemes exhibit a gradual increase in required E_b/N_0, but m-CAP maintains a 2–3 dB advantage, particularly at high data rates. This improvement stems from m-CAP's spectrally efficient sub-band modulation and pulse shaping, which better exploit the usable portion of the LED bandwidth. DCO-OFDM, although effective due to its multicarrier structure, is less power-efficient and requires higher SNR to meet the same BER target.

In summary, M-PPM is ideal for ultra-low-rate CubeSat links (up to 16 Mbps) due to its excellent energy efficiency. OOK is best suited for a narrow midrate range (16–20 Mbps), where it outperforms both M-PPM and m-CAP. For higher data rates, m-CAP offers the best performance, combining spectral efficiency and lower power requirements, making it a strong candidate for high-throughput intersatellite optical links.

Taken together, these results define quantitative operating regions under the assumed 20 MHz LED and noise model: M-PPM is viable and energy-optimal only up to approximately 16 Mbps, OOK occupies a narrow but useful window around 16–20 Mbps, and beyond this range m–CAP becomes the only scheme that simultaneously satisfies the BER target and maintains a practical E_b/N_0 budget. This transforms the discussion from qualitative expectations ("PPM for low rate, OFDM for high rate") into explicit design boundaries in the $(R_b, E_b/N_0)$ domain that can directly guide CubeSat optical link configuration.

D. Required Transmit Power Versus Communication Range Under Daytime and Nighttime Conditions

Fig. 5 illustrates the required transmit power P_t as a function of ISC range under both daytime and nighttime background conditions. The analysis targets a BER of 10^{-3} for three modulation schemes chosen according to their respective optimal data rate regimes: M-PPM at 1 Mbps, OOK at 20 Mbps, and m-CAP at 70 Mbps. As demonstrated in Sections IV-B and IV-C, DCO-OFDM is systematically outperformed by m-CAP in the high-data-rate regime, with a consistent penalty of approximately 2–3 dB. Consequently, the corresponding DCO-OFDM curve follows the same distance-dependent trend as m-CAP, but with a systematic upward offset in the required transmit power. Including this curve would therefore be redundant and would not provide additional physical insight beyond what is already conveyed by the m-CAP result. In addition, note that the ambient irradiance is set to 11.8 W/m^2 for daytime and 2.0 W/m^2 for nighttime, reflecting the significant variation in solar background noise.

Across all distances and data rates, daytime operation requires significantly more transmit power—typically close to double that of nighttime conditions. This is due to increased background-induced shot noise at the receiver under solar illumination. The influence of background light highlights the critical role of environmental conditions in the design of CubeSat optical intersatellite links.

The results also validate the data-rate-based selection of modulation schemes. Each modulation demonstrates optimal power efficiency within its designated regime. For instance, M-PPM is clearly the most power-efficient at low data rates. At 1 km, it requires only 113.3 mW during nighttime and 232.4 mW during the daytime. This makes it ideal for low-throughput, long-duration CubeSat missions where energy constraints are stringent.

OOK offers a good tradeoff for moderate data rates. At 1 km, it requires 138.1 mW at night and 295 mW during the day—only about 20% more than M-PPM, despite supporting a 20× higher throughput. This makes OOK suitable for telemetry, command data, and moderate-resolution payload transfers, balancing power efficiency and system simplicity.

m-CAP is used for high-throughput links and offers the best spectral efficiency, albeit with higher power demands. At 1 km, it requires 438.5 mW at night and 899.4 mW during daytime. While more power-hungry, it enables three times the data rate of OOK. Its use is well-justified for short-to-medium-range intersatellite links, especially when downlinking large datasets or relaying high-bandwidth payload data between satellites.

Two additional insights emerge from this analysis. First, even in the worst-case scenario (m-CAP at 70 Mbps during daytime at 1 km), the required transmit power remains below 900 mW. This is within the power budget of most CubeSats, especially when transmissions are duty-cycled or occur in short bursts. Second, the 2× power saving during nighttime strongly supports mission planning strategies that favor high-data-rate communications when satellites are oriented away from direct solar exposure. Such temporal optimization can substantially reduce the energy burden on CubeSats, improving mission sustainability without sacrificing performance.

In summary, the selected modulation schemes are well-aligned with their intended data rate ranges, and the required power levels remain feasible within CubeSat hardware constraints. Environmental factors—particularly solar background irradiance—have a substantial effect on system performance, emphasizing the need for power-aware and context-adaptive communication planning in LED-based ISC system design.

It is important to note that the results in Fig. 5 correspond to a realistic nonideal operating point with $\eta_p = 0.8$ and $\eta_{\text{CSI}} = 0.9$. To further quantify the impact of CSI and pointing imperfections, Table III reports the required transmit power at 1 km for independent values of η_p and η_{CSI}, including the ideal case. In accordance with the analytical scaling law $P_t \propto 1/(\eta_p^2 \eta_{\text{CSI}})$, impairments introduce a systematic power offset while preserving the same distance-dependent trend observed in Fig. 5.

For OOK at 20 Mbps (nighttime), the required transmit power increases from 81.1 mW under ideal conditions to 111.3 mW for moderate impairments ($\eta_p = \eta_{\text{CSI}} = 0.9$) and to 138.1 mW for the Fig. 5 operating point, before reaching 158.5 mW under severe conditions ($\eta_p = \eta_{\text{CSI}} = 0.8$). A similar behavior is observed for m-CAP at 70 Mbps (daytime), where the power requirement rises from 528.2 (ideal) to 724.6, 899.4, and 1031.7 mW, respectively.

These results confirm a predictable performance degradation under increasing CSI and pointing uncertainties. Importantly, even under severe impairments, the required transmit power remains compatible with typical CubeSat power budgets, thereby validating the robustness of the proposed intersatellite optical communication framework.

TABLE III
Impact of Independent Pointing and CSI Efficiencies on Required Transmit Power At 1 km (PIN Receiver)

Case (1 km)	η_p	η_{CSI}	P_t (mW)
OOK (20 Mbps, Night)	1.0	1.0	81.1
OOK (20 Mbps, Night)	0.9	0.9	111.3
OOK (20 Mbps, Night)	0.8	0.9	138.1
OOK (20 Mbps, Night)	0.8	0.8	158.5
m-CAP (70 Mbps, Day)	1.0	1.0	528.2
m-CAP (70 Mbps, Day)	0.9	0.9	724.6
m-CAP (70 Mbps, Day)	0.8	0.9	899.4
m-CAP (70 Mbps, Day)	0.8	0.8	1031.7
M-PPM (1 Mbps, Night)	1.0	1.0	66.5
M-PPM (1 Mbps, Night)	0.9	0.9	91.3
M-PPM (1 Mbps, Night)	0.8	0.9	113.3
M-PPM (1 Mbps, Night)	0.8	0.8	130.0

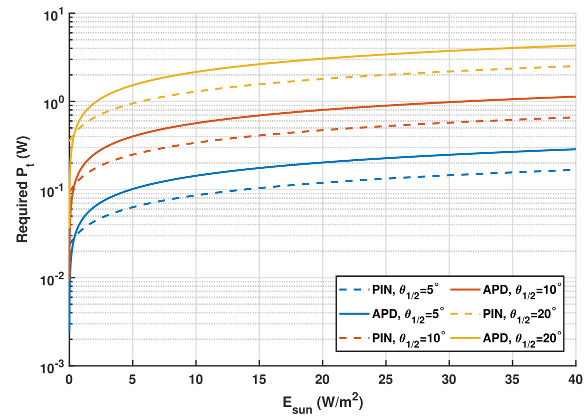


Fig. 6. Required transmit power P_t vs. solar irradiance E_{sun} for PIN and APD detectors at different LED half-power semiangles ($\theta_{1/2} = 5^\circ, 10^\circ, 20^\circ$), with $R_b = 20$ Mbps and fixed receiver FOV $\psi_c = 5^\circ$. Solid lines represent APD detectors, and dashed lines represent PIN detectors.

E. Impact of Background Noise on BER Performance and Required Transmit Power

To assess system robustness under varying ambient illumination, we evaluate the required transmit power P_t for PIN and APD detectors across different receiver FOV angles (ψ_c) and LED half-power semiangles ($\theta_{1/2}$). The analysis assumes an OOK modulation scheme operating at a communication distance of 500 m and a data rate $R_b = 20$ Mbps, targeting a BER of 10^{-3}.

Fig. 6 presents the required transmit power P_t as a function of background noise power P_{bkg} for different LED half-power semiangles ($\theta_{1/2} = 5^\circ, 10^\circ, 20^\circ$), with the receiver FOV fixed at 5°. Narrower beams reduce geometric spreading loss and significantly lower the required P_t, but at the expense of stricter pointing accuracy requirements.

The comparison between PIN and APD detectors further reveals key differences in performance under varying background noise levels. At low background noise levels, APDs offer a power advantage due to their internal gain. However, this benefit is limited to low-noise regimes. Specifically, APDs outperform PIN detectors only when the background noise is below approximately $P_{\mathrm{bkg}} = 0.5\ \mathrm{W/m^2}$.

At nighttime irradiance levels ($P_{\mathrm{bkg}} = 2\ \mathrm{W/m^2}$), PIN detectors actually require less transmit power than APDs for all tested LED divergence angles. For instance, with a half-power semiangle of $\theta_{1/2} = 10^\circ$, the required transmit power for PIN is 180.4 mW, compared to 262.6 mW for APD. As background noise increases to daytime levels ($P_{\mathrm{bkg}} = 10.81\ \mathrm{W/m^2}$), this difference becomes more pronounced: the required power for PIN reaches 471.2 mW, while APDs require 798.8 mW under the same conditions. This behavior stems from the excess noise inherent in APD operation, which scales significantly with background-induced shot noise, and it is obtained under deliberately optimistic APD assumptions. Furthermore, the assumptions made regarding APD performance are particularly optimistic, as they consider an active detection area of 1 $\mathrm{cm^2}$—an unusually large size that, to our knowledge, is not commercially available. Implementing such a configuration would therefore necessitate a complex detector design, potentially involving multiple APDs coupled with parallel transimpedance amplifiers (TIAs). As a result, the APD performance curves in this article should be interpreted as a best-case upper bound; using a more realistic, smaller APD area would only increase the required APD transmit power and thus further strengthen the conclusion that PIN photodiodes are more suitable for the considered CubeSat LED links.

Although APDs are often favored in low-photon environments due to their internal gain, the results show that PIN detectors deliver more consistent and power-efficient performance under high-background conditions, such as during sunlit orbital passes. Their stable and predictable response to background noise makes them well-suited for continuously illuminated CubeSat missions.

Given these advantages and typical CubeSat constraints, PIN detectors are recommended for LED-based intersatellite links. When combined with narrow FOV and beam divergence angles (e.g., 5°), they offer strong background noise rejection and improved link efficiency. For typical intersatellite distances ranging from 10 to 1000 m, this configuration yields a beam footprint of approximately 87 at 1000 m—well within the pointing capabilities of modern CubeSat attitude control systems, which routinely achieve subdegree accuracy.

In summary, PIN detectors paired with moderate optical alignment parameters present a robust and energy-efficient solution for LEO intersatellite optical links, balancing noise resilience, power consumption, and alignment feasibility.

V. CONCLUSION

This article detailed the performance analysis of LED-based OWC systems for ISC in CubeSat constellations. By incorporating realistic LED bandwidth limitations, background illumination from solar, lunar, and terrestrial sources, and key optical link design parameters, we evaluated the feasibility of power- and bandwidth-constrained optical systems under stringent CubeSat constraints.

Our results show that PIN photodetectors consistently outperform APDs under typical LEO conditions, particularly when solar irradiance exceeds 0.5 $\mathrm{W/m^2}$. For instance, under daylight conditions ($P_{\mathrm{bkg}} = 10.81\ \mathrm{W/m^2}$), PIN-based receivers require up to 30% less transmit power than APD-based counterparts to maintain a BER below 10^{-3}. Narrow LED semiangles (e.g., $\theta_{1/2} = 5^\circ$) and tight receiver FOVs (e.g., $\psi_c = 5^\circ$) were shown to significantly suppress background noise and enhance link robustness, though at the cost of increased alignment sensitivity. With beam footprints below 90 m at 1 km range, these configurations remain feasible within the capabilities of current CubeSat attitude control systems.

We introduced and validated analytical BER expressions for four key modulation schemes—OOK, M-PPM, DCO-OFDM, and m-CAP—capturing the tradeoffs between complexity, spectral efficiency, and power requirements. A detailed evaluation of their BER performance under realistic LED bandwidth constraints and varying data rates revealed distinct operating regimes for each scheme. Beyond qualitative expectations, the proposed models enabled quantitative identification of the precise data-rate boundaries at which each scheme remains viable under the same hardware and noise conditions. M-PPM emerged as the most energy-efficient choice at ultra-low data rates (up to 16 Mbps), while OOK offered superior performance in the intermediate range (16–20 Mbps). Above this range, LED-induced ISI causes OOK to become increasingly power-inefficient, while m-CAP systematically outperforms DCO-OFDM by 2–3 dB due to its subband pulse shaping and more efficient exploitation of the usable LED spectrum. Additionally, an assessment of required transmit power versus communication range under both daytime and nighttime irradiance conditions confirmed that all three schemes operate within feasible power budgets (below 400 mW at 1 km). These results translate into concrete design rules for CubeSat optical links: M-PPM for energy-constrained telemetry, OOK for narrow midrate windows, and m-CAP for high-throughput exchanges.

Future research should focus on refining the irradiance and noise models to account for time-varying orbital dynamics, satellite attitude fluctuations, and shadowing effects. Incorporating spatiotemporal variability will enhance predictive accuracy under operational conditions. Moreover, advancing beam alignment through closed-loop tracking systems or adaptive optics will be essential for sustaining narrow-beam links, especially over long ranges. Exploring hybrid modulations that combine the robustness of PPM with the spectral efficiency of m-CAP or OFDM could unlock further performance gains. Finally, machine learning-driven adaptation techniques hold promise for optimizing link parameters and resource allocation in real time, particularly in dense, dynamic CubeSat constellations where interference and variability are significant.

In conclusion, this work provides a comprehensive framework for the design and optimization of LED-based ISC systems tailored to CubeSat platforms. By integrating refined analytical BER models with realistic hardware constraints and environmental conditions, the article delivers nontrivial quantitative operating boundaries and system-level guidelines that are directly applicable to future CubeSat optical architectures.

APPENDIX A
DERIVATION OF BER FOR BANDWIDTH-LIMITED OOK

This appendix derives the approximate BER expression (28) for OOK when the LED is bandwidth-limited and when the receiver noise includes both physical noise and intersymbol interference (ISI). The development uses a received-signal formulation based on the effective channel gain $\tilde{H}(0)$, which includes geometric loss, concentrator and FOV effects, pointing loss, and CSI estimation efficiency.

In OOK, a bit "1" is transmitted using a rectangular pulse of amplitude i_{peak}, while a bit "0" is transmitted with zero current. Modeling the LED as a first-order low-pass filter with time constant τ, the output current for a transmitted "1" is

$$s_{\text{out}}(t) = i_{\text{peak}}\left(1 - e^{-t/\tau}\right), \qquad 0 \le t \le T_b. \tag{47}$$

Averaging over one bit duration yields the effective peak current

$$\begin{aligned} i_{\text{peak,eff}} &= \frac{1}{T_b}\int_0^{T_b} s_{\text{out}}(t)\,dt \\ &= \frac{i_{\text{peak}}}{T_b}\int_0^{T_b}\left(1 - e^{-t/\tau}\right)dt \\ &= i_{\text{peak}}\left[1 - \frac{\tau}{T_b}\left(1 - e^{-T_b/\tau}\right)\right]. \end{aligned} \tag{48}$$

Let $\tilde{H}(0)$ denote the effective channel DC gain, including geometric loss, optical concentrator effects, FOV terms, pointing loss, and CSI efficiency. If $H(0)$ is the LOS gain, η_p is the pointing efficiency, and η_{CSI} the CSI efficiency, then

$$\tilde{H}(0) = \sqrt{\eta_{\text{CSI}}}\,\eta_p\,H(0). \tag{49}$$

The received optical power and noise-free photocurrent are therefore

$$P_r(t) = \kappa\,\tilde{H}(0)\,s_{\text{out}}(t) \tag{50}$$

$$i_r(t) = R\,\kappa\,\tilde{H}(0)\,s_{\text{out}}(t) \tag{51}$$

where κ is the electro–optical efficiency and R the detector responsivity.

The average received current for an ideal (undistorted) "1" pulse is

$$I_{r,1}^{\text{ideal}} = R\,\kappa\,\tilde{H}(0)\,i_{\text{peak}} \tag{52}$$

while under LED bandwidth limitation, the average becomes

$$I_{r,1} = R\,\kappa\,\tilde{H}(0)\,i_{\text{peak,eff}}. \tag{53}$$

Because of LED memory, a "1" in the previous bit induces a residual tail current in the next bit. Assuming the previous bit is "1" with probability 1/2, the tail waveform in the next bit is

$$s_{\text{tail}}(t) = i_{\text{peak}}e^{-t/\tau}, \qquad 0 \le t < T_b. \tag{54}$$

The corresponding average residual electrical current is

$$\begin{aligned} I_{\text{ISI}} &= \frac{1}{2T_b}\int_0^{T_b} R\,\kappa\,\tilde{H}(0)\,i_{\text{peak}}e^{-t/\tau}\,dt \\ &= \frac{R\,\kappa\,\tilde{H}(0)\,i_{\text{peak}}}{2}\frac{\tau}{T_b}(1 - e^{-T_b/\tau}) \\ &= \frac{\tau}{T_b}\left(\frac{I_{r,1}^{\text{ideal}}}{2}\right)(1 - e^{-T_b/\tau}). \end{aligned} \tag{55}$$

This ISI current contributes an additional shot-noise term with variance

$$\sigma_{\text{ISI}}^2 \approx 2q\,I_{\text{ISI}}\,B \tag{56}$$

leading to a total decision variance

$$\sigma_{\text{tot,eff}}^2 = \sigma_{\text{tot}}^2 + \sigma_{\text{ISI}}^2 = \sigma_{\text{tot}}^2(1 + \delta_{\text{ISI}}) \tag{57}$$

with

$$\delta_{\text{ISI}} = \frac{\sigma_{\text{ISI}}^2}{\sigma_{\text{tot}}^2}. \tag{58}$$

Thus

$$Y|0 \sim \mathcal{N}(0, \sigma_{\text{tot,eff}}^2), \qquad Y|1 \sim \mathcal{N}(I_{r,1}, \sigma_{\text{tot,eff}}^2). \tag{59}$$

Using the standard midpoint threshold

$$\eta_r = \frac{I_{r,1}^{\text{ideal}}}{2} \tag{60}$$

the conditional error probabilities are

$$P(e|0) = Q\left(\frac{\eta_r}{\sigma_{\text{tot,eff}}}\right), \qquad P(e|1) = Q\left(\frac{I_{r,1} - \eta_r}{\sigma_{\text{tot,eff}}}\right). \tag{61}$$

Define the ISI-free SNR

$$\text{SNR}_0 = \frac{(I_{r,1}^{\text{ideal}})^2}{4\sigma_{\text{tot}}^2} = \frac{(R\kappa\tilde{H}(0)i_{\text{peak}})^2}{4\sigma_{\text{tot}}^2}. \tag{62}$$

Using $\sigma_{\text{tot,eff}}^2 = \sigma_{\text{tot}}^2(1 + \delta_{\text{ISI}})$, the effective OOK SNR becomes

$$\text{SNR}_{\text{eff,OOK}} = \frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}. \tag{63}$$

The Q-function arguments then read

$$\frac{\eta_r}{\sigma_{\text{tot,eff}}} = \sqrt{\frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}} \tag{64}$$

$$\frac{I_{r,1} - \eta_r}{\sigma_{\text{tot,eff}}} = 2\sqrt{\frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}}\left[\frac{1}{2} - \frac{\tau}{T_b}(1 - e^{-T_b/\tau})\right]. \tag{65}$$

Finally, for equally likely bits, the BER is

$$P_{be,\text{OOK}}^{\text{eff}} = \frac{1}{2}Q\left(\sqrt{\frac{\text{SNR}_0}{1 + \delta_{\text{ISI}}}}\right)$$

$$+\frac{1}{2}Q\left(2\sqrt{\frac{\mathrm{SNR}_0}{1+\delta_{\mathrm{ISI}}}}\left[\frac{1}{2}-\frac{\tau}{T_b}(1-e^{-T_b/\tau})\right]\right). \tag{66}$$

In the limit $\tau \to 0$ and $\delta_{\mathrm{ISI}} \to 0$, the LED becomes memoryless and the classical OOK BER expression under AWGN is recovered with SNR_0.

APPENDIX B
BUSSGANG-BASED DERIVATION OF THE DCO–OFDM EFFECTIVE SNR

This appendix derives the effective SNR expression used in Section III-C for biased and clipped DCO–OFDM. The derivation uses a Bussgang decomposition applied to the IFFT output of the Hermitian-symmetric QAM vector, and the resulting mapping links the nominal OOK-based SNR SNR_0 to the per-bit SNR of DCO–OFDM under IM/DD constraints.

Let $x[n]$ denote the real-valued IFFT output in (35). For a large number of active subcarriers, the central limit theorem ensures that $x[n]$ is well approximated by a zero-mean Gaussian random variable with variance σ_x^2, i.e.,

$$x[n] \sim \mathcal{N}(0, \sigma_x^2). \tag{67}$$

To satisfy the IM/DD nonnegativity constraint, a DC bias B_{DC} is added and all negative values are clipped, resulting in the DCO–OFDM waveform

$$s_{\mathrm{DCO}}[n] = \max\{x[n] + B_{\mathrm{DC}}, 0\}. \tag{68}$$

Defining the normalized bias $\alpha = B_{\mathrm{DC}}/\sigma_x$ and writing $x[n] = \sigma_x Z$ with $Z \sim \mathcal{N}(0,1)$, one obtains

$$s_{\mathrm{DCO}}[n] = (\sigma_x Z + B_{\mathrm{DC}})\,\mathbf{1}_{\{Z>-\alpha\}}. \tag{69}$$

Following Bussgang's theorem, the biased–clipped waveform can be decomposed into a scaled linear component plus a distortion term uncorrelated with $x[n]$

$$s_{\mathrm{DCO}}[n] = G_{\mathrm{c}}\, x[n] + d[n] + B_{\mathrm{eff}} \tag{70}$$

where G_{c} is the Bussgang gain, $d[n]$ is the clipping distortion, and B_{eff} absorbs the mean DC level. Since distortion is uncorrelated with $x[n]$, the useful and distortion powers are

$$P_{\mathrm{sig}} = G_{\mathrm{c}}^2\sigma_x^2, \qquad P_{\mathrm{clip}} = \mathbb{E}\{|d[n]|^2\} \tag{71}$$

and the total average power at the LED input satisfies

$$P_{\mathrm{avg}} = P_{\mathrm{sig}} + P_{\mathrm{clip}}. \tag{72}$$

For biased–clipped Gaussian inputs, the Bussgang gain simplifies to

$$G_{\mathrm{c}} = \Phi(\alpha) \tag{73}$$

so that

$$P_{\mathrm{sig}} = \Phi(\alpha)^2\sigma_x^2. \tag{74}$$

Using standard Gaussian integrals, the total waveform power can be expressed as

$$P_{\mathrm{avg}} = \sigma_x^2\big(\Phi(\alpha) - \alpha\phi(\alpha)\big) + 2\sigma_x B_{\mathrm{DC}}\phi(\alpha) + B_{\mathrm{DC}}^2\Phi(\alpha) \tag{75}$$

where $\phi(\alpha)$ and $\Phi(\alpha)$ are the Gaussian pdf and cdf, respectively.

Let the received electrical signal be

$$y[n] = R\kappa\tilde{H}(0)\, s_{\mathrm{DCO}}[n] + n[n] \tag{76}$$

with noise variance σ_{tot}^2. Using the Bussgang decomposition, the time-domain SNR becomes

$$\mathrm{SNR}_{\mathrm{time}} = \frac{P_{\mathrm{sig}}}{P_{\mathrm{avg}}} \cdot \frac{\mathrm{SNR}_0}{1 + \frac{P_{\mathrm{clip}}}{P_{\mathrm{avg}}}\mathrm{SNR}_0}. \tag{77}$$

DCO–OFDM suffers additional losses due to cyclic prefix overhead, the removal of DC and Nyquist subcarriers, and the conversion from symbol-level to bit-level SNR. Collecting these factors yields the effective per-bit SNR

$$\mathrm{SNR}_{\mathrm{eff,DCO}} = \mathrm{SNR}_0\frac{P_{\mathrm{sig}}}{P_{\mathrm{avg}}}\left[1 + \frac{P_{\mathrm{clip}}}{P_{\mathrm{avg}}}\mathrm{SNR}_0\right]^{-1} \times \frac{N_{\mathrm{FFT}} - 2}{N_{\mathrm{FFT}} + N_{\mathrm{cp}}} \cdot \frac{1}{\log_2 M_{\mathrm{QAM}}}. \tag{78}$$

In the operating regime of interest (moderate bias, negligible upper clipping), the ratio $P_{\mathrm{sig}}/P_{\mathrm{avg}}$ is well approximated by

$$\frac{P_{\mathrm{sig}}}{P_{\mathrm{avg}}} \approx \frac{1}{1+\gamma}, \qquad \gamma = \frac{B_{\mathrm{DC}}^2}{\sigma_x^2} \tag{79}$$

leading directly to the compact expression used in (37).

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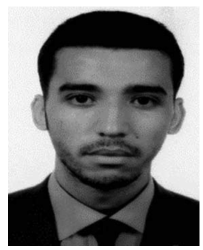

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